

EETS 7302 TCP/IP Network Administration

Session 3

Network Layer (IP): quick review

SWITCHING

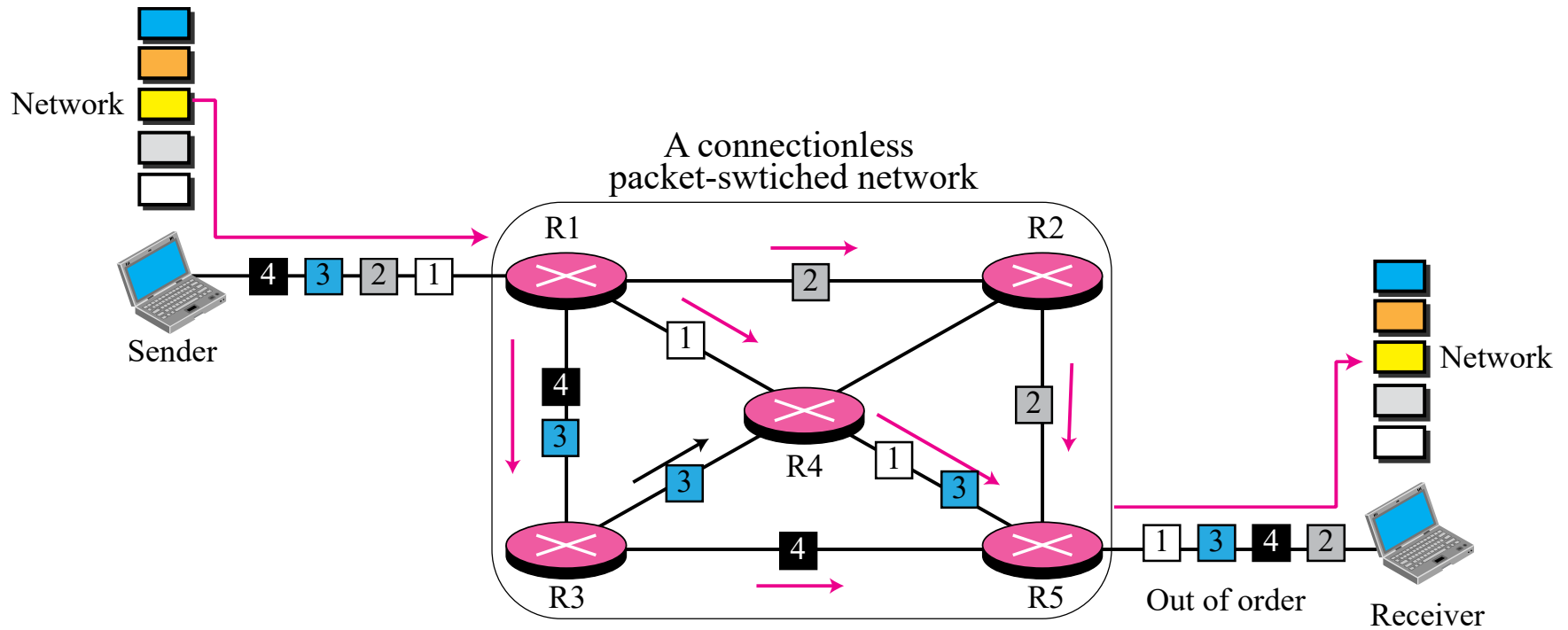
The connecting device acts as a switch that connects one port to another port.

- ✓ **Circuit Switching**
- ✓ **Packet Switching**

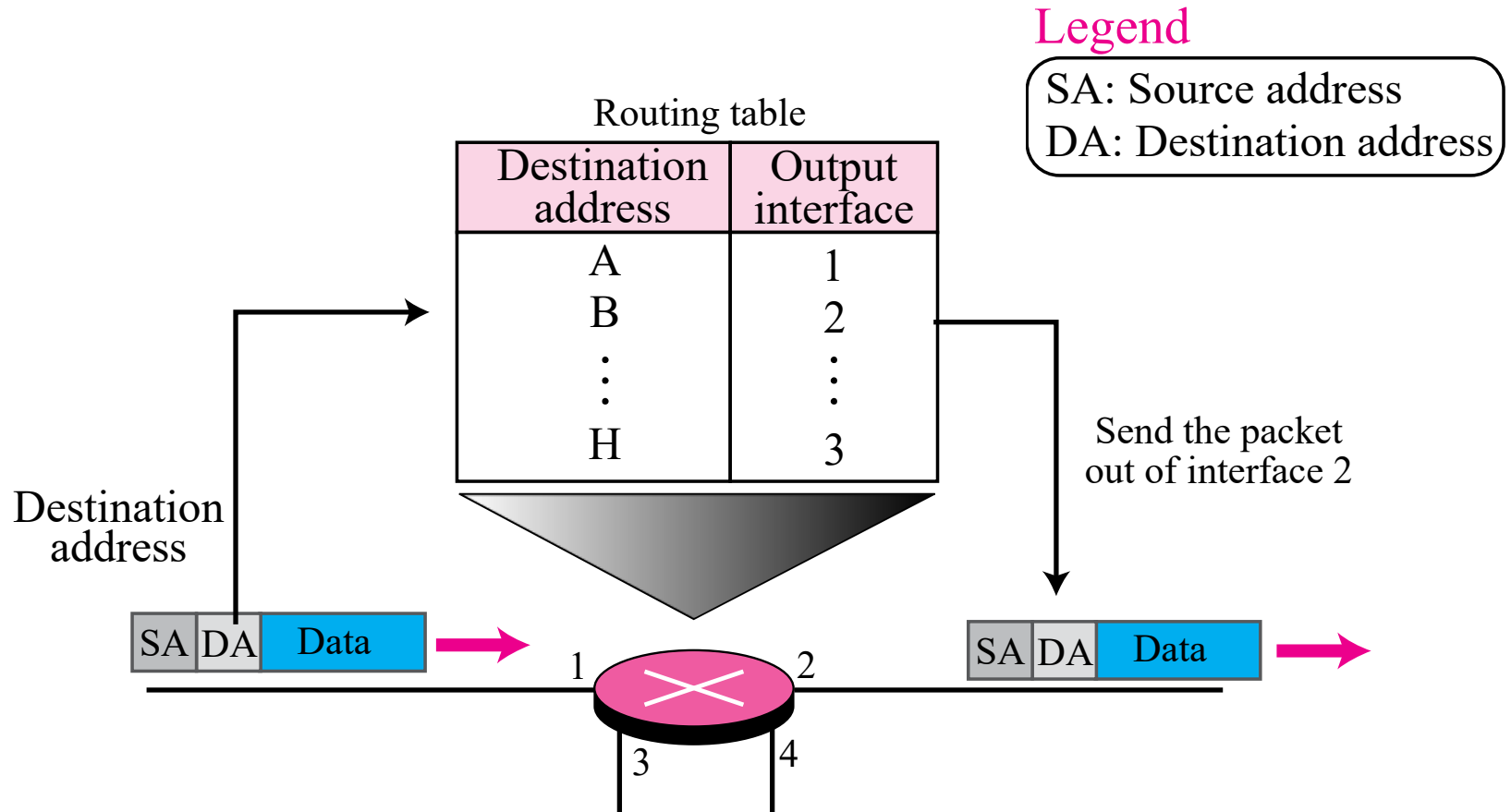
PACKET SWITCHING

- The network layer is designed as a packet-switched network.
- Message is divided into manageable packets
- Individual datagrams are then transferred from the source to the destination.
- The received datagrams are assembled at the destination
 - connectionless service,
 - connection-oriented service.

A connectionless packet-switched network

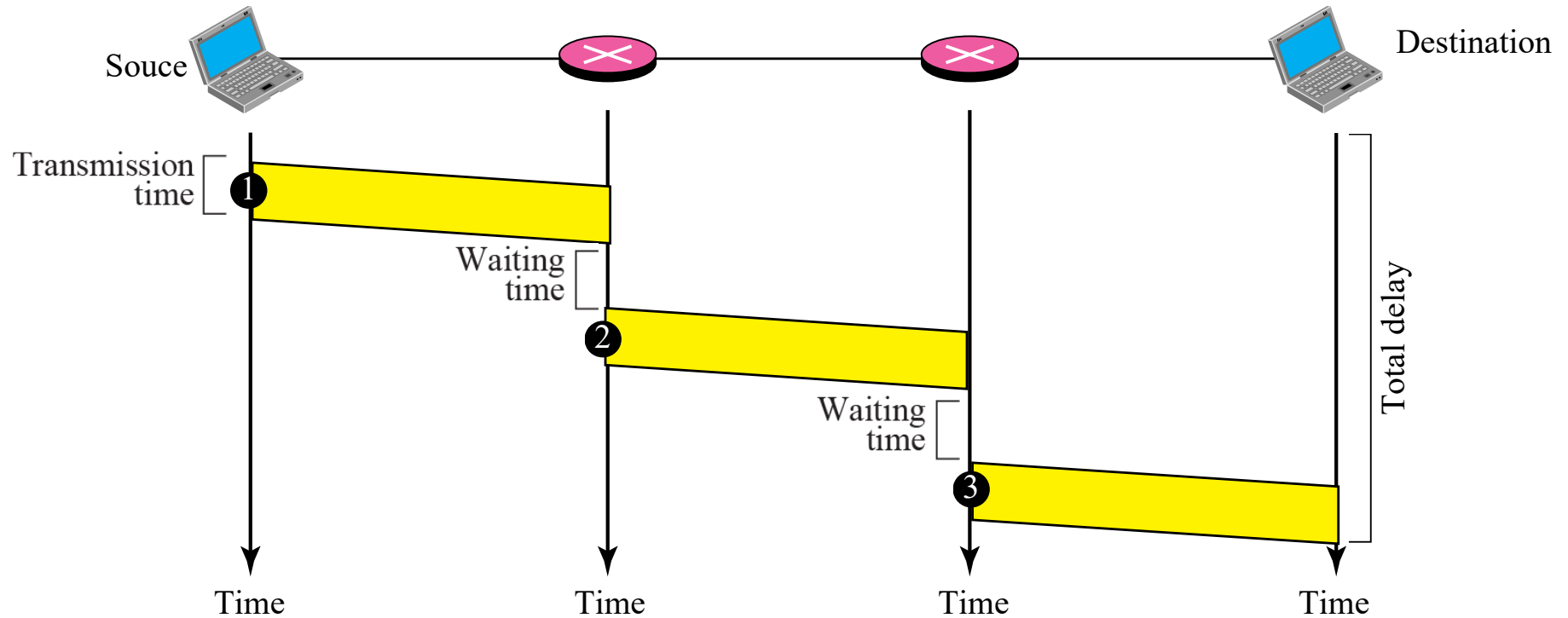


Forwarding process in a connectionless network

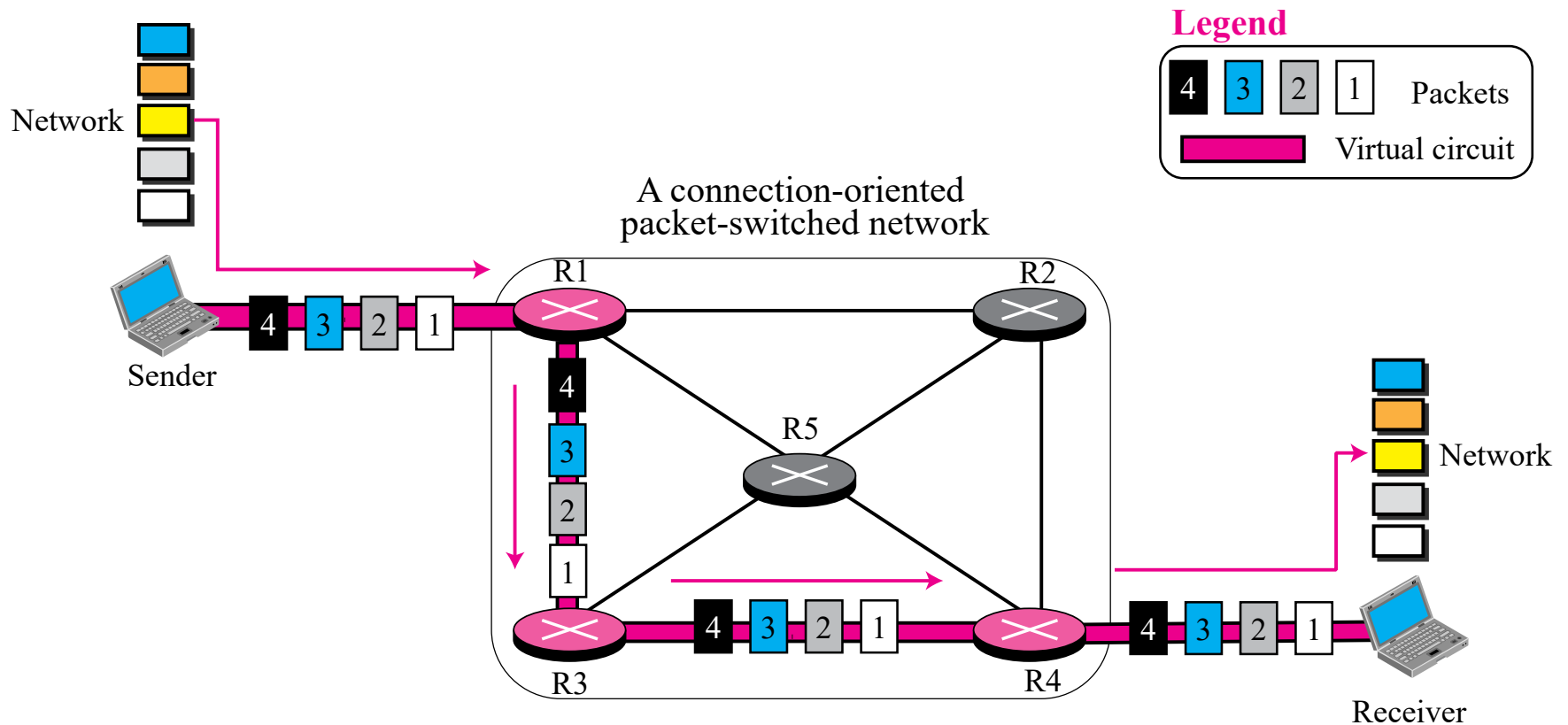


In a connectionless packet-switched network, the forwarding decision is based on the destination address of the packet.

Delay in a connectionless network

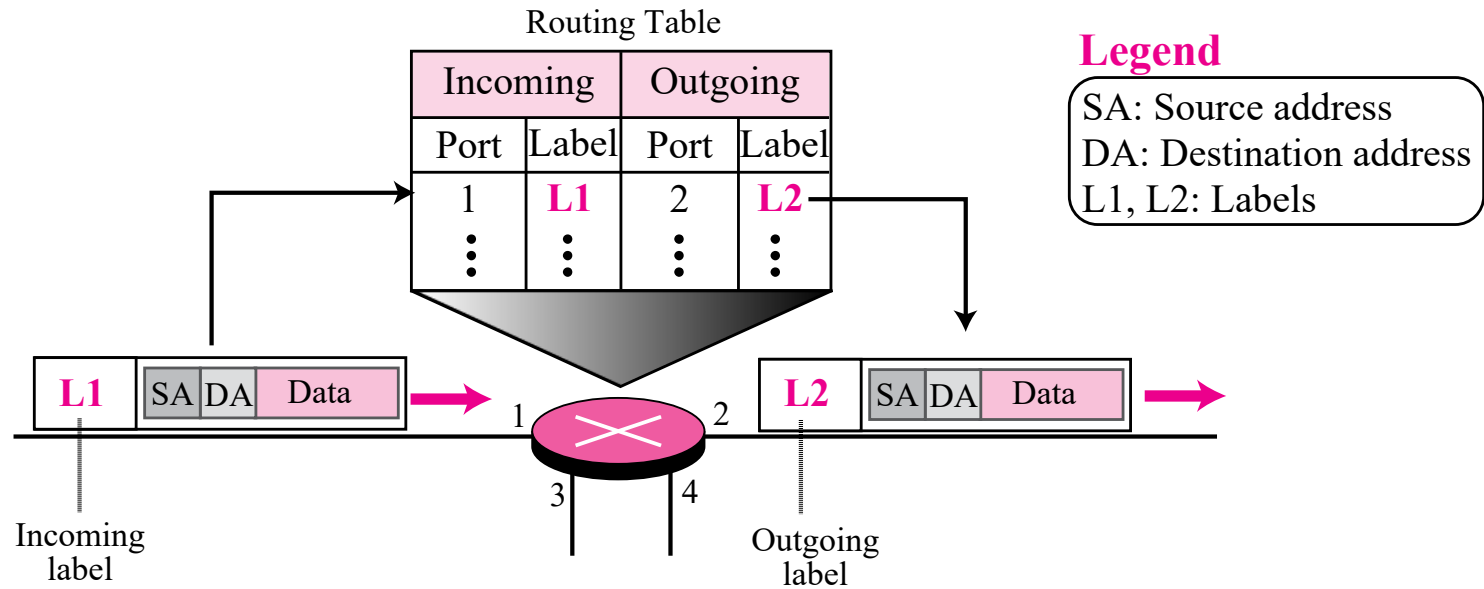


A connection-oriented packet switched network

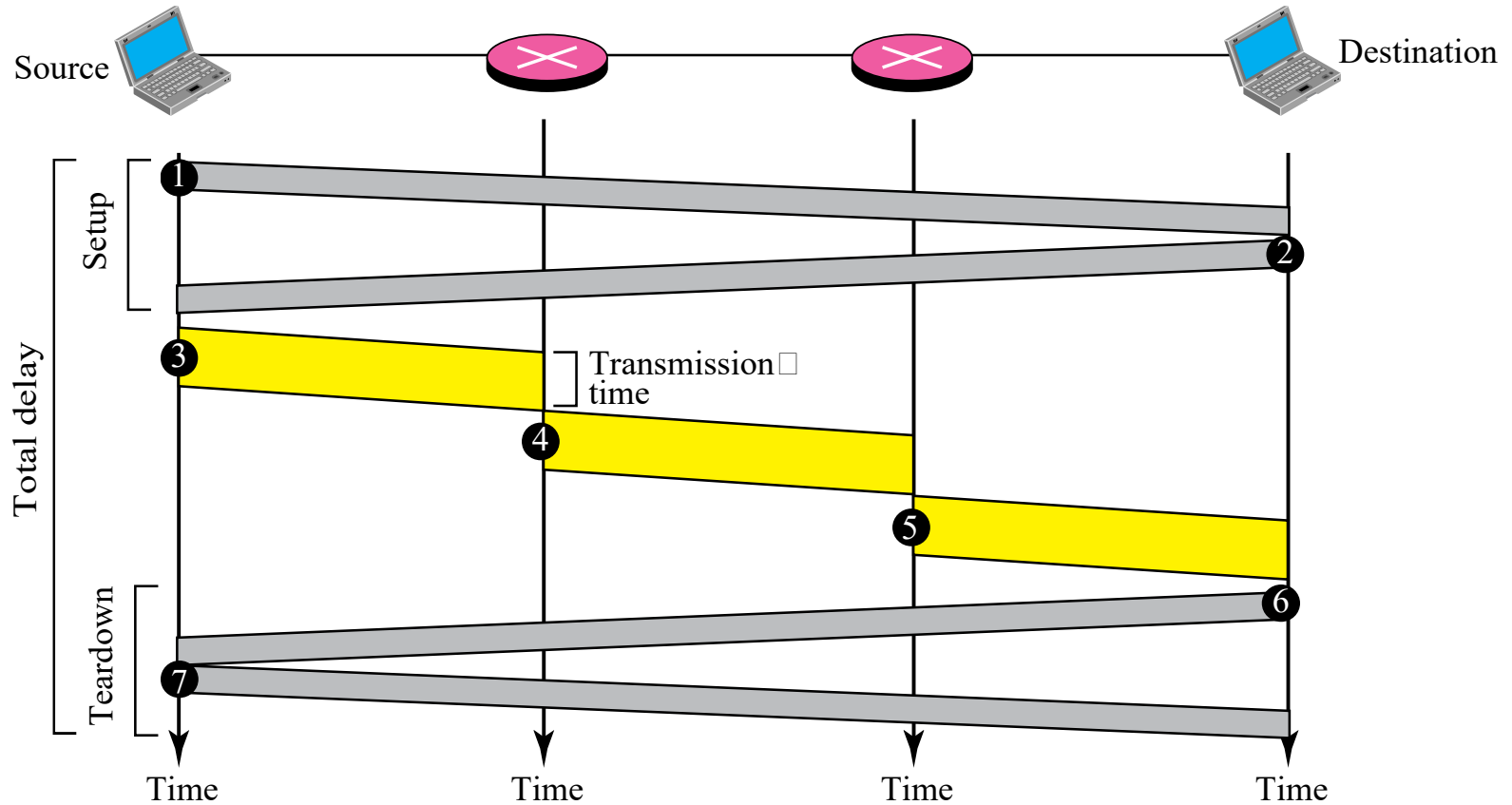


In a connection-oriented packet switched network, the forwarding decision is based on the label of the packet.

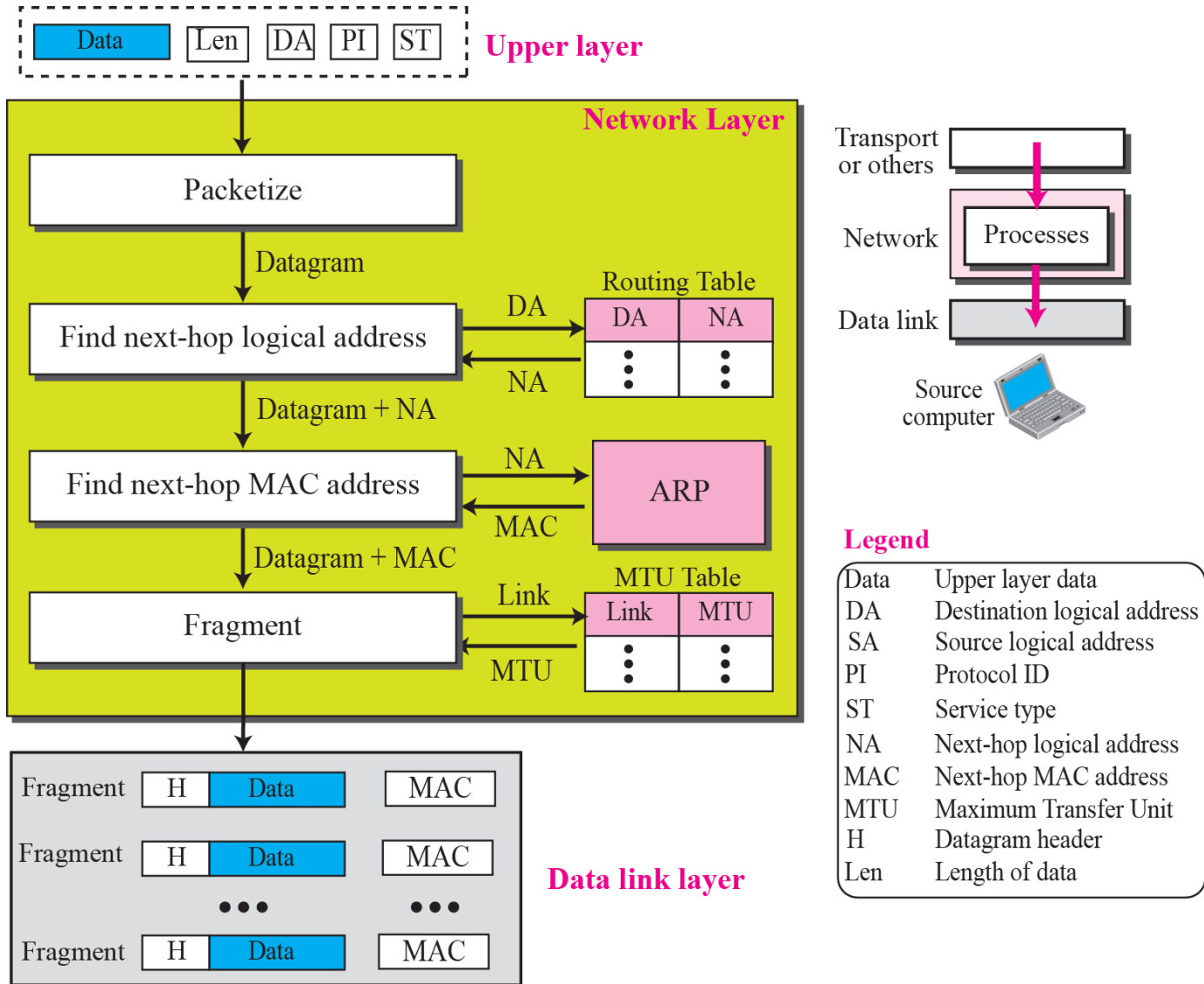
Forwarding process in a connection-oriented network



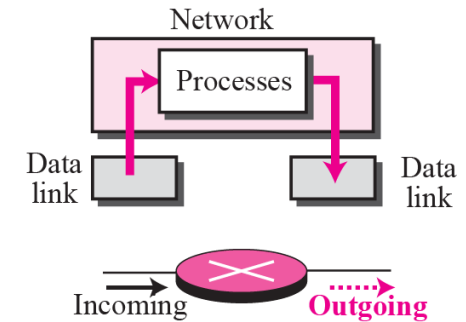
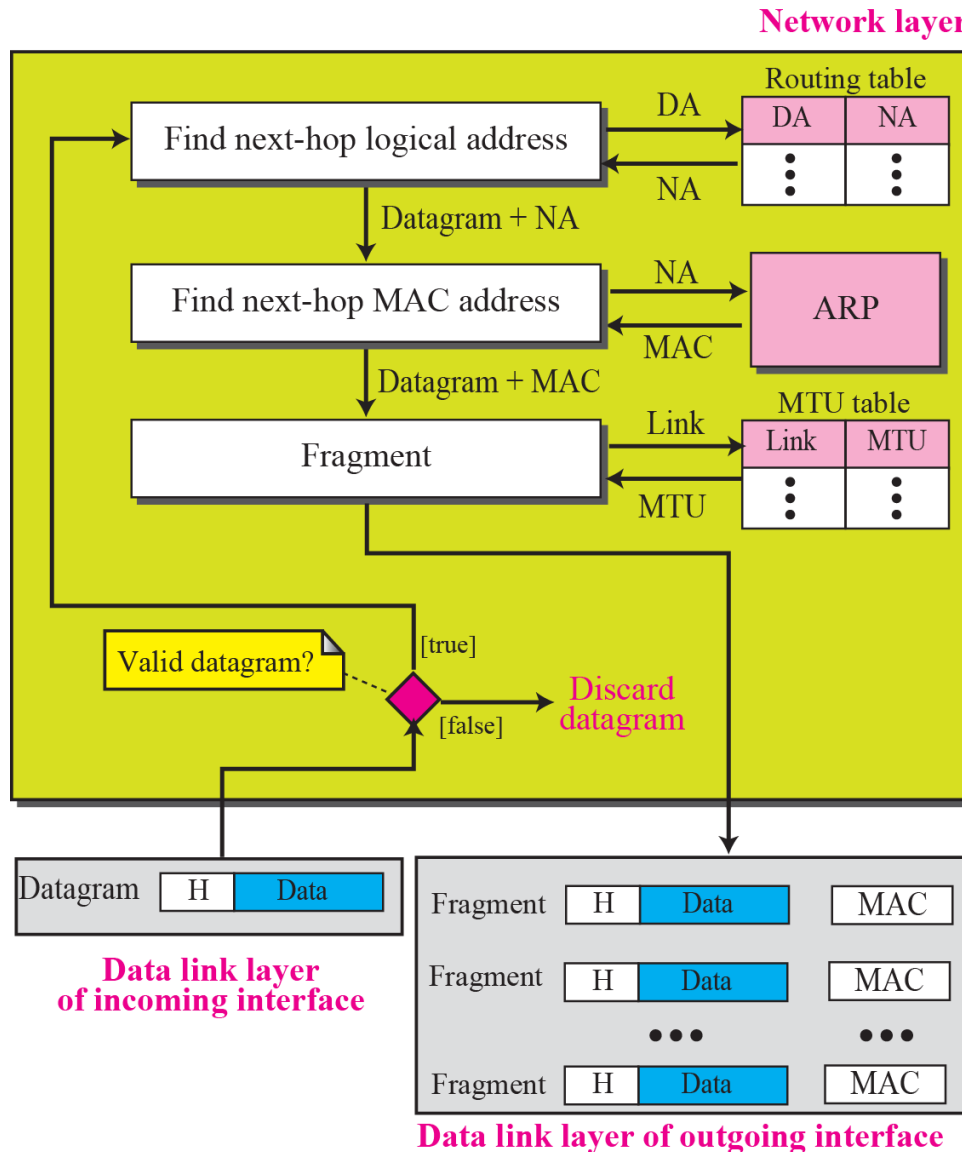
Delay in a connection-oriented network



Network Layer Services provided at the source computer



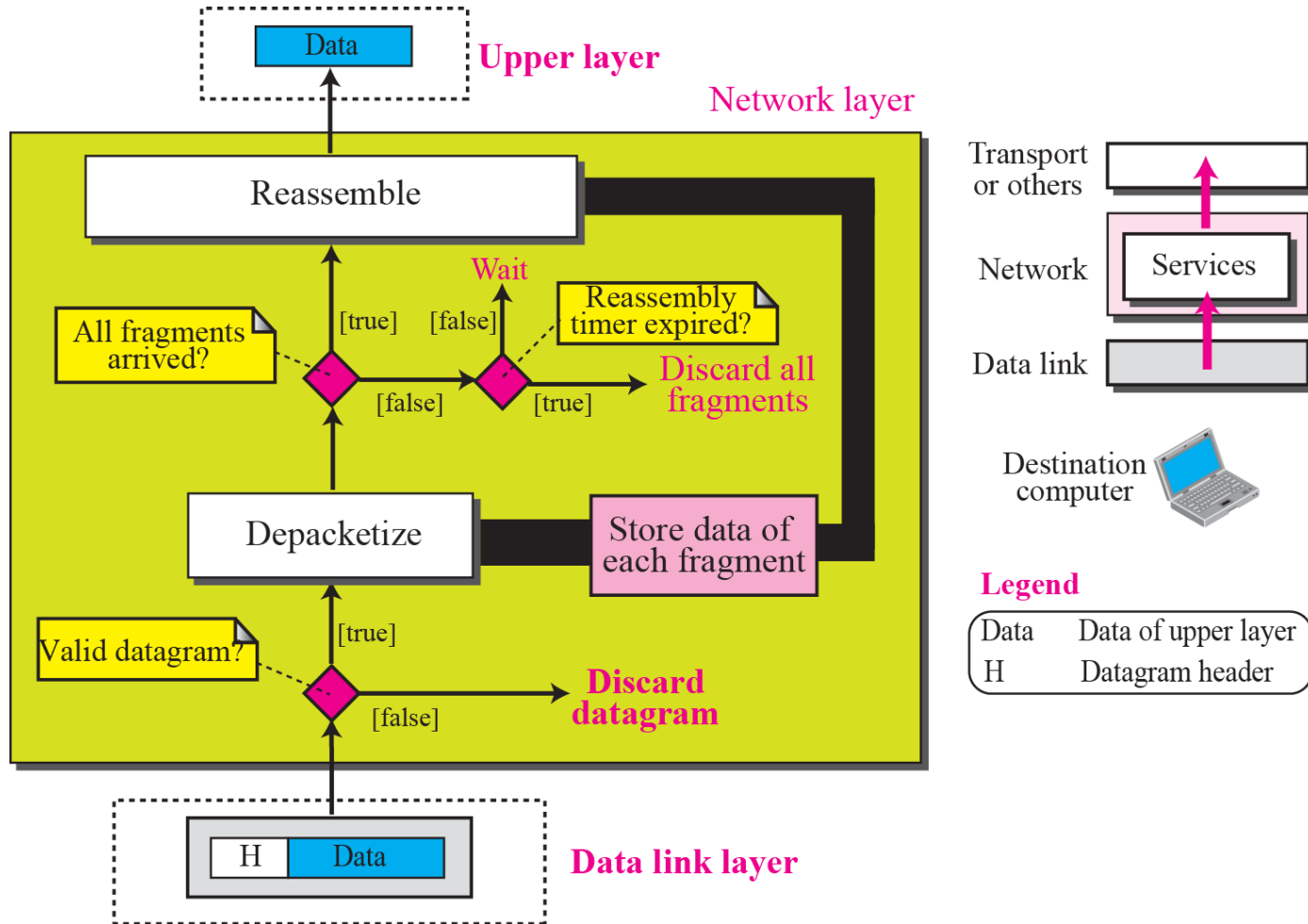
Network Layer Processing at each router



Legend

Data	Upper layer data
DA	Destination logical address
NA	Next-hop logical address
MAC	Next-hop MAC address
MTU	Maximum Transfer Unit
H	Datagram header

Network Layer Processing at the destination computer



References

**TCP/IP Protocol Suite (4th Edition),
by Behrouz A. Forouzan – Used the text's slide deck**

End of session